Algorithmic Redistricting and Ensemble Methods:
A Framework for Comparison

Establishing the G-Track Methodology

G.0 — Ensemble Methodology

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May 2026

Abstract

Markov-chain ensemble methods and algorithmic redistricting pursue fundamentally different goals: the former characterises the space of "reasonable" plans in order to *evaluate* any given plan against a geographic baseline; the latter produces a single canonical plan that is verifiably optimal by a stated criterion. These goals are complementary but often conflated in legal and academic discourse. This paper establishes the methodological framework for the G-series, which situates the ApportionRegions algorithm within the ReCom ensemble to quantify where the algorithmic plan falls along every relevant distribution—Polsby-Popper compactness, partisan seat share, and minority representation. We review GerryChain, ReCom, and Short-Burst methods; introduce the percentile framework for locating a single plan in an ensemble distribution; connect ensemble convergence diagnostics ($\hat{R}$, ESS, Hamming autocorrelation) to the ConvergenceSweep stopping criterion established in U.1; and articulate the statutory implications of a *verifiable deterministic plan* against 10,000 *unverifiable stochastic*

plans. The G-track papers (G.1–G.5) instantiate this framework on six states with published ensemble data.

Contents

1. Introduction

Markov-chain ensemble methods have become the dominant tool for detecting gerrymandering in U.S. congressional redistricting. The core idea, introduced by Chikina et al. [2017] and refined by DeFord et al. [2021], is to generate thousands of "neutral" maps by random walk on the space of valid districting plans, then ask: where does the contested map fall in the resulting distribution? If the map lies in the 99th percentile of partisan skew, one infers that geography alone cannot explain the skew, and that intentional partisan sorting is the likely cause.

This inference engine is powerful precisely because it is comparative. The ensemble does not tell you what the *correct* map is. It tells you whether a given map is an outlier relative to what geography permits. Courts have begun to rely on this logic: ensemble evidence appeared in *League of Women Voters of Pennsylvania v. Commonwealth of Pennsylvania* (2018), *Common Cause v. Rucho* (2019), and was implicitly acknowledged in *Rucho v. Common Cause* even as the majority declined to apply a federal standard.

The ApportionRegions algorithm occupies a different position in this landscape. Rather than sampling the space to characterise it, ApportionRegions directly computes the minimum-edge-cut partition of the state's census-tract graph under the prime factorization of the seat count. This is not a sampled plan—it is a *local optimum* under a specific objective function, reached deterministically from a census-derived seed via ConvergenceSweep [Della-Libera, 2026b].

Two different goals. We insist on a clean distinction:

> **Goal A (Ensemble as Evaluator)**: Generate a distribution of "reasonable" plans from the geographic constraints alone. Use this distribution to evaluate any single plan—partisan, court-drawn, or algorithmic—by its percentile position.
>
> **Goal B (Algorithm as Canonical Plan)**: Compute a single plan that is optimal by a stated criterion (minimum edge cut, maximum compactness, etc.) and verifiably so. The plan is canonical—reproducible from public data by anyone.

These goals are complementary: the ensemble provides the reference distribution against which the canonical plan's properties can be quantified. But they are not substitutes. An ensemble plan is stochastic and unverifiable in the sense that no particular sampled plan has a privileged claim. The canonical plan is deterministic and verifiable: given the 2020 census data and the seat count, there is exactly one ApportionRegions map for each state.

The G-track. This paper establishes the methodology for the G-series of companion papers. G.1 computes the percentile position of the ApportionRegions plan within published ensemble results for six states. G.2 focuses on partisan outcome distributions. G.3 examines compactness distributions. G.4 develops convergence diagnostics ($\hat{R}$, ESS, Hamming autocorrelation). G.5 addresses mixing analysis for the ReCom chain itself.

Organisation. Section 2 reviews ensemble methods. Section 3 introduces the percentile framework. Section 4 covers convergence diagnostics. Section 5 connects the diagnostics to ConvergenceSweep. Section 6 discusses statutory implications. Section 7 concludes.

2. Ensemble Methods Overview

We review the three principal methods used to generate redistricting ensembles, focusing on their outputs, limitations, and what each method is suited to establish.

2.1. GerryChain and the Flip Chain

GerryChain [MGGG Redistricting Lab, 2021] implements a Markov chain on the space of valid k-district plans for a given state graph. The basic transition is a *flip*: a randomly chosen boundary node is reassigned to an adjacent district, provided the resulting plan remains connected and within population tolerance. The chain is run for millions of steps; at each step (or at a thinned subset of steps) the current plan is recorded.

Outputs. The flip chain produces a sample from the *stationary distribution* of valid plans. If run long enough and sampled frequently enough, the histogram of any statistic (partisan seats, Polsby-Popper score, minority district count) approximates the true geometric distribution of that statistic over all valid plans.

Limitations.

- **Mixing time.** The flip chain mixes slowly on large state graphs (e.g., Texas at $k = 38$). Published ensemble sizes of 10,000 to 1,000,000 plans may not achieve stationarity.

- **Locality bias.** Flips are local moves. The chain may not explore topologically distinct map families within feasible runtime.

- **No canonical plan.** Any sampled plan is one draw from the distribution. No plan in the ensemble has a privileged status.

2.2. ReCom (Recombination)

ReCom [DeFord et al., 2021] replaces the flip transition with a *recombination* move: two adjacent districts are merged, then the merged region is re-partitioned by a random spanning tree cut. This produces much larger structural changes per step.

Outputs. ReCom generates plans with different properties than the flip chain. In particular, ReCom tends to produce more "wiggly" district boundaries, while the flip chain tends toward more blocky shapes. Both are valid samples from their respective stationary distributions, but these distributions are not identical.

Limitations.

- **Non-reversibility.** Standard ReCom is not reversible with respect to a known target distribution. The stationary distribution is implicit.

- **Compactness underweighting.** The random spanning tree cut does not weight for compactness, so ReCom ensembles often contain plans less compact than what a human mapmaker would draw.

- **Different distribution from flip chain.** Results from ReCom and flip-chain ensembles cannot be directly compared without calibration.

The Autry et al. [2021] Metropolized Multiscale Forest ReCom partially addresses the non-reversibility issue by applying a Metropolis-Hastings correction, but at substantial computational cost.

2.3. Short-Burst Methods

Short bursts [Cannon et al., 2022] are a variant of ReCom designed specifically to optimise a target objective. The chain is run for a short fixed number of steps (a "burst"), the best plan in the burst is kept, and the process repeats. Short bursts can hill-climb toward extreme plans (e.g., maximum minority-district count) while remaining within the convex hull of valid plans.

Outputs. Short-burst ensembles are not samples from the geographic distribution. They are samples from the optimum-seeking distribution around a specific objective. They are useful for establishing what is *achievable* under the geographic constraints, not what is *typical*.

Limitations.

- Not a characterisation of the full plan space.
- Results depend strongly on the objective function and burst length.
- Cannot be used to place a plan in the "neutral" distribution.

2.4. What Each Method Establishes

Table 1 summarises the three methods.

Table 1: Comparison of ensemble methods and their evidentiary uses.

Method	Stationary dist. known	Compactness weight	Best evidentiary use
Flip chain	Implicit (uniform)	None	Outlier detection
ReCom	Implicit	None	Partisan distribution
Short-burst	None (optimiser)	None	Feasibility bounds
ApportionRegions (this work)	N/A (deterministic)	METIS edge cut	Canonical plan

The ApportionRegions plan does not belong to any of these categories: it is not a sample, but the output of a deterministic optimisation. The G-track papers use the ReCom ensemble as the reference distribution and locate the ApportionRegions plan within it.

3. The Algorithmic Plan's Position in the Ensemble

3.1. Percentile Definition

Let $\mathcal{E} = \{P_1, P_2, \ldots, P_N\}$ be an ensemble of N valid redistricting plans for a state with k districts. Let $f : \mathcal{P} \to \mathbb{R}$ be a scalar statistic (e.g., Polsby-Popper mean, Democratic seat share, minority-majority district count). The *percentile* of the algorithmic plan P^* with respect to f and $\mathcal{E}$ is:

$$\pi_f(P^*) = \frac{1}{N}\#\{i \in [N] : f(P_i) \leq f(P^*)\}. \tag{1}$$

A plan at the 50th percentile is typical; plans at the 5th or 95th percentile are unusual but not necessarily extreme. Courts have generally treated plans above the 99th or below the 1st percentile as requiring explanation [Herschlag et al., 2020].

3.2. Statistics of Interest

The G-track considers three families of statistics:

1. **Polsby-Popper (PP) compactness.** For a district with area A and perimeter P, $\mathrm{PP} = 4\pi A/P^2 \in (0, 1]$. Higher is more compact. We report the plan-level mean $\bar{\mathrm{PP}}$ and the minimum $\mathrm{PP}_{\min}$ across districts.

2. **Partisan seat share.** The number of districts won by each party under a specified electoral baseline (e.g., 2020 presidential vote). We report Democratic seats $S_D \in \{0, 1, \ldots, k\}$.

3. **Minority-majority districts (MM).** The count of districts with minority voting-age population exceeding 50%. We report $\mathrm{MM} \in \{0, 1, \ldots, k\}$.

3.3. Interpreting Percentile Position

The percentile framework supports two distinct claims:

Claim 1 (Compactness not extreme). If $\pi_{\bar{\mathrm{PP}}}(P^*) \in [0.60, 0.90]$, the ApportionRegions plan is more compact than average but not an outlier. This is *legally desirable*: a plan at the 99th percentile of compactness might be suspected of packing communities to achieve partisan ends via the "compactness route."

Claim 2 (Partisanship near median). If $\pi_{S_D}(P^*) \in [0.40, 0.60]$, the partisan outcome of the ApportionRegions plan is indistinguishable from the geographic median. This is the strongest possible evidence that the plan is not gerrymandered.

We will see in G.1 and G.2 that ApportionRegions achieves both claims for the majority of studied states, with notable exceptions in states with strong geographic partisan sorting (Georgia, Texas).

3.4. Sensitivity to Ensemble Choice

The percentile $\pi_f(P^*)$ depends on the choice of ensemble $\mathcal{E}$. Different chain types (flip vs. ReCom) produce different distributions; different ensemble sizes affect statistical precision; different population tolerance bounds affect plan diversity. In G.1, we use published ensemble results from Herschlag et al. [2020] for North Carolina and DeFord et al. [2021] for other states, acknowledging that these are ReCom ensembles with $N \geq 10{,}000$ and population tolerance $\pm 0.5\%$.

Where published ensembles are unavailable, we use literature-reported distributional summaries (mean, standard deviation, approximate percentile bounds) to bound $\pi_f(P^*)$. These bounds are marked with † throughout the G-track papers.

3.5. Package Evidence Boundary

The percentile in Equation 1 is an evidence claim only when both sides of the comparison are archived. A complete G-track package must bind the external trace, the deterministic bisect plan, the RCTX context, the metric output, and any election-input model used for partisan seats. The current repository records two complementary fixtures:

- `docs/examples/g-ensemble-evidence-packages/G.1-G.3+active-synthetic/` is an active synthetic package that exercises every required artifact role.
- `docs/examples/g-ensemble-evidence-packages/G.1-G.3+missing-evidence/` records the remaining real external-trace, election, metric, diagnostic, RPLAN, and RCTX gaps for headline G.1–G.3 empirical percentiles.

Thus, G.0 should be read as the methodology contract. The synthetic package shows that the contract is executable; real state percentile claims require real packages with the same roles.

4. Convergence Diagnostics

Before using an ensemble for inference, one must verify that the Markov chain has approximately converged to its stationary distribution. We review three diagnostics used in the G-track and developed in detail in G.4.

4.1. Gelman-Rubin $\hat{R}$

The Gelman and Rubin [1992] potential scale reduction factor is computed from $m \geq 2$ independent chains of length n. Let $\bar{\psi}_{j\cdot}$ be the within-chain mean and $\bar{\psi}_{\cdot\cdot}$ the grand mean of statistic ψ:

$$B = \frac{n}{m-1} \sum_{j=1}^{m} (\bar{\psi}_{j\cdot} - \bar{\psi}_{\cdot\cdot})^2 \quad \text{(between-chain variance)} \tag{2}$$

$$W = \frac{1}{m} \sum_{j=1}^{m} s_j^2 \quad \text{(within-chain variance)} \tag{3}$$

$$\hat{V} = \frac{n-1}{n} W + \frac{m+1}{mn} B \tag{4}$$

$$\hat{R} = \sqrt{\hat{V}/W}. \tag{5}$$

A value $\hat{R} < 1.1$ is conventionally accepted as indicating convergence for redistricting's discrete integer outcomes.[1] For redistricting ensembles, we compute $\hat{R}$ over partisan seats and Polsby-Popper means.

4.2. Effective Sample Size (ESS)

The effective sample size corrects for autocorrelation within a chain. For a chain of length N with autocorrelation ρ_t at lag t:

$$\text{ESS} = \frac{N}{1 + 2\sum_{t=1}^{\infty} \rho_t}. \tag{6}$$

An ESS of at least 400 per statistic is the minimum floor for reliable 95% interval estimates; G.4 [**?**] establishes 500 as the recommended target based on empirical diagnostics across six

[1]Vehtari et al. (2021) recommend $\hat{R} < 1.01$ for continuous parameters; 1.1 is the appropriate threshold for the discrete seat-count and plan-assignment statistics used here.

states. ReCom ensembles typically achieve higher ESS than flip-chain ensembles of the same length, because the recombination move decorrelates faster.

4.3. Hamming Autocorrelation

District-level statistics like seat share depend on which tracts belong to which district, not just on continuous quantities. The *Hamming distance* between two plans P and P' is:

$$d_H(P, P') = \frac{1}{n}\#\{i : \text{district}(t_i, P) \neq \text{district}(t_i, P')\}, \tag{7}$$

where t_i ranges over census tracts and we match districts by population overlap. The *Hamming autocorrelation* at lag ℓ is:

$$\rho_H(\ell) = \frac{\text{Cov}[d_H(P_t, P_{\text{ref}}), d_H(P_{t+\ell}, P_{\text{ref}})]}{\text{Var}[d_H(P_t, P_{\text{ref}})]}, \tag{8}$$

where P_{ref} is a fixed reference plan. Rapid decay of $\rho_H(\ell)$ to zero indicates that the chain is exploring the plan space efficiently. G.4 reports Hamming autocorrelation plots for all six states.

4.4. Diagnostic Standards for the G-Track

The following standards apply across all G-track papers:

Diagnostic	Threshold	Consequence of failure
$\hat{R}$	< 1.1	Results flagged as preliminary
ESS	> 500 (min 400)	Wider uncertainty intervals used
$\rho_H(10)$	< 0.10	Chain considered mixed

For published ensembles (Herschlag 2020, DeFord 2021) we reproduce the authors' reported convergence diagnostics. Where these are unavailable, we apply conservative uncertainty adjustments to reported percentiles.

4.5. Diagnostic Artifacts

Diagnostic thresholds are not self-authenticating. A package that cites $\hat{R}$, ESS, or Hamming autocorrelation must include the trace or summary from which those quantities were computed, the seed policy, the chain count, the burn-in/thinning rule, and the statistic definitions. In the package vocabulary used by the repository, these are `external-trace` and `diagnostic` artifacts. Missing either artifact downgrades a percentile statement from empirical finding to protocol or design claim.

The active synthetic G.1–G.3 package validates this artifact shape at small scale. The missing-evidence package remains the live ledger for real diagnostic artifacts that must be added before G.1–G.3 percentiles can be cited as completed external-ensemble findings.

5. The ConvergenceSweep Bridge

5.1. Two Notions of Convergence

The term "convergence" is used differently in the ensemble and algorithmic settings. For ensemble methods, convergence refers to the Markov chain approaching its stationary distribution. For ConvergenceSweep [Della-Libera, 2026b], convergence refers to the optimisation search reaching the minimum edge-cut map for a given state and seat count.

These are distinct phenomena, but they interact. The key question is:

> At the threshold $T = 600$, ConvergenceSweep produces the minimum-edge-cut (ApportionRegions) plan. This plan is a *specific point* in the plan space that the ReCom chain also explores. Where does this specific point fall in the ReCom distribution?

5.2. Characterising the Local Optimum

The ApportionRegions plan is a local optimum of the METIS edge-cut objective with respect to small perturbations. Under ReCom moves, the plan will not remain in its basin of attraction for long—a single recombination move typically changes 5–15% of district assignments and can escape local optima. This means the ApportionRegions plan is *not* the typical plan in the ReCom ensemble.

However, the ApportionRegions plan's *compactness* (as measured by Polsby-Popper score) is empirically high: T.4 shows a mean $\bar{\mathrm{PP}} \approx 0.38$–$0.45$ across states, versus ReCom ensemble means of 0.28–0.35. This places the ApportionRegions plan typically at the 65th–75th percentile of the ReCom compactness distribution—more compact than average but not extreme.

5.3. The Minimum-EC Plan in the ReCom Distribution

Let P^*_{EC} denote the minimum-edge-cut plan found by ConvergenceSweep. The edge-cut objective $\mathrm{EC}(P)$ is negatively correlated with Polsby-Popper compactness (lower edge cut implies fewer shared-boundary census tracts across district lines, which correlates with more cohesive district shapes). Within the ReCom ensemble, plans with lower EC are rarer, because ReCom does not target EC minimisation.

Empirically, we estimate P^*_{EC} falls at approximately the

- 65th–75th percentile of the $\bar{\mathrm{PP}}$ distribution,
- 45th–60th percentile of the S_D (partisan seats) distribution,
- 50th–65th percentile of the MM distribution,

with state-specific variation characterised in G.1–G.3. The partisan percentile near the median is the central finding: the ApportionRegions plan's partisanship is close to what geography determines, not to what a partisan mapmaker would achieve.

5.4. Diagnostic Consistency

The $T = 600$ threshold for ConvergenceSweep was certified in U.1 using empirical data from B.7 (50-state seed sweep).[2] The same data supports ensemble diagnostic thresholds: if the seed sweep finds no improvement after 600 seeds, the plan is stable under seed perturbation. This is analogous to—but distinct from—an ensemble chain achieving ESS > 500 (recommended; minimum floor is 400).

ConvergenceSweep $T = 600$ and ensemble ESS are independent certificates of robustness: ESS measures MCMC mixing in the ensemble space; $T = 600$ measures monotonic convergence in the seed-quality space. Both certify stability but via orthogonal mechanisms. They are not numerically comparable and should not be conflated. Together they provide complete coverage of the two-method framework: ESS certifies that the ensemble distribution estimate is stable under additional draws; $T = 600$ certifies that the deterministic search has found the best available plan under additional seeds.

[2] The T=600 ConvergenceSweep formula is certified in U.1 **?** (Round 2, accepted; mean reviewer score 3.67/4). The formula's theoretical foundations and 89-seed empirical margin above Georgia's certified tail are documented in that paper.

6. Statutory Implications

6.1. One Verifiable Plan vs. 10,000 Unverifiable Plans

Ensemble evidence is powerful for *detection* but weak for *selection*. A court can use an ensemble to determine that a contested plan is an outlier. But which plan should replace it? The ensemble offers 10,000 candidates, none of which has a principled claim over the others. A special master tasked with drawing a remedial map cannot simply "pick the median ensemble plan" without introducing a new layer of discretion.

The ApportionRegions plan solves this problem. Given the census data and the seat count, there is exactly one ApportionRegions plan per state. It is:

- **Verifiable**: Anyone with access to the public census data and the `bisect` binary can reproduce it exactly, bit-for-bit.
- **Deterministic**: No human judgment enters after the inputs are fixed.
- **Politically inert**: No partisan registration data, no electoral history, no demographic targets beyond population equality.
- **Constitutionally grounded**: The ApportionRegions algorithm derives from Article I §2 via the prime factorization of the Huntington-Hill seat count [Della-Libera, 2026a].

An ensemble plan cannot make the first three claims. Even if a court selected the 50th-percentile plan on compactness, that plan reflects one random draw and would change with a different seed, a different run, or a different ensemble construction.

6.2. The *Rucho v. Common Cause* Framework and Algorithmic Plans

Rucho v. Common Cause held that partisan gerrymandering claims present a political question not cognizable in federal court. The majority distinguished between identifying a gerrymander (possible) and providing a judicially manageable standard for remediation (not possible under the Constitution). The dissent argued that ensemble evidence provides exactly such a standard.

The ApportionRegions plan offers a different response: bypass the manageability problem entirely by providing a plan that requires no judicial discretion to select. The court does not choose among plans; it certifies the ApportionRegions plan as constitutionally required by the Huntington-Hill apportionment process. This argument does not depend on *Rucho v. Common Cause* being overruled; it operates at the Article I §2 level rather than the Equal Protection or First Amendment level.

6.3. Ensemble Evidence as Corroboration

Within the G-track framework, ensemble evidence serves as *corroboration* for the ApportionRegions plan rather than as its foundation. The argument structure is:

1. The ApportionRegions plan is constitutionally required (T.4, B.02).
2. The ApportionRegions plan is not a partisan outlier: its partisan percentile is near the ensemble median (G.1, G.2).
3. The ApportionRegions plan is compact but not an extreme outlier on compactness: its compactness percentile is 65–75th (G.3).
4. The ApportionRegions plan converges reproducibly: ConvergenceSweep at $T = 600$ certifies optimality (U.1).

Claims 2 and 3 are corroborative: they show that the ApportionRegions plan possesses properties independently associated with non-gerrymandered maps. They do not constitute the primary legal argument. This ordering is important: ensemble evidence is used to verify that the deterministic plan is reasonable, not to select the plan.

7. Conclusion

The G-track establishes a rigorous framework for situating the ApportionRegions algorithmic redistricting plan within the ensemble landscape. The central methodological contribution is the distinction between two goals: ensemble methods characterise the space of reasonable plans (Goal A), while the ApportionRegions algorithm computes the canonical optimal plan within that space (Goal B).

These goals are not competitors. The ensemble distribution is the natural reference frame for evaluating the ApportionRegions plan. If ApportionRegions produces a plan that falls at the 54th percentile of partisan outcomes and the 68th percentile of compactness, the ensemble has confirmed that the plan is neither a partisan outlier nor a geometric curiosity. The plan can then be presented to a court or legislature with two certificates: it is the unique minimum-edge-cut plan for this state's geography and seat count, and it lies in the interior of the space of geographically reasonable plans.

The convergence diagnostics from G.4 ($\hat{R}$, ESS, Hamming autocorrelation) apply to both ensemble and algorithmic validation. The ConvergenceSweep threshold $T = 600$ from U.1 translates into an analogue of ESS for the seed sweep: stability across 600 consecutive non-improving seeds is evidence that the seed space has been adequately explored.

The statutory implication is direct. Courts seeking a remedial plan that does not require judicial discretion can adopt the ApportionRegions plan. Ensemble evidence, as developed in G.1–G.3, confirms that the plan is geographically reasonable on the three dimensions courts have examined: compactness, partisan balance, and minority representation. The plan is not perfect on any dimension—it is optimal on the minimum-edge-cut dimension—but it is defensible on all dimensions, which is what the law requires.

Future work in the G-track (G.4, G.5) will develop ensemble diagnostics for the ReCom chain itself and examine mixing rates as a function of state complexity (number of tracts, k, geographic compactness). These papers will also provide the computational foundation for running bespoke ensembles where published results are unavailable.

The current implementation state is more precise than the original G-track framing. The repository now contains an executable synthetic package showing the artifact contract for G.1–G.3, and a separate missing-evidence package identifying the real artifacts still needed for empirical percentile claims. Accordingly, G.0's methodological claim is package-shaped: ensemble comparisons are publication-ready only when trace, diagnostic, metric, election, RPLAN, and RCTX artifacts are hash-bound and replayable.

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